Cost-Sensitive Awareness-Based SAR Automatic Target Recognition for Imbalanced Data

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***Abstract*— With the maturity of synthetic aperture radar (SAR) technology, the problem of imbalanced data has gradually emerged. This problem makes it difficult for the automatic target recognition (ATR) model to properly learn the classification boundaries of majority and minority category target samples. In this article, we propose an ATR model with new architecture, called the cost-sensitive awareness-based automatic target recognition (CA-ATR) model, which provides an effective way of solving the problem of imbalanced data. Aimed at the two issues caused by imbalanced data on ATR models, the proposed method solves the problems from both the data and algorithm levels. At the data level, CA-ATR avoids adverse correlations among the target samples through different oversampling methods. By making the ATR model cost-sensitive, the proposed method also avoids the empirical risk preference of the ATR model for majority category target samples at the algorithm-level. At the same time, CA-ATR can autonomously learn different cost-sensitive awareness from different imbalanced data sets. The awareness enables the ATR model to more accurately learn the classification boundaries between target samples that belong in different categories. Several experimental results show the superiority of the proposed approach based on the Moving and Stationary Target Acquisition and Recognition (MSTAR) data set. Compared with other imbalanced learning methods, the proposed method is able to solve different types of imbalanced data problems.**

***Index Terms*— Automatic target recognition (ATR), cost-sensitive awareness, imbalanced data, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an important method for earth observation sensors using remote sensing. SAR, a high-resolution imaging radar, has all-weather, day, and night operational capabilities, which far outmatches the performance capabilities of other sensors [1]. One of the most important applications of SAR is the automatic target recognition (ATR) technology, which is a critical technology in the fields of military and homeland security [2]–[4]. ATR technology has continued to significantly improve over the past decade, due to the rapid development of machine learning algorithms.

Among the many ATR technologies, the data-driven SAR ATR algorithms contribute the most to the improvement of recognition performance. The use of data-driven SAR ATR algorithms has a long history, and its development can be traced back through extensive literature. The common feature extraction algorithms, such as principal component analysis (PCA) [5]–[7], nonnegative matrix factorization (NMF) [8], and linear discriminant analysis (LDA) [9], [10], seek better data representation in target samples. In addition, the support vector machine (SVM) maximizes the classification margin between different categories of target samples by selecting support vectors, achieving great success in ATR technology [11], [12]. Moreover, deep learning algorithms have received more and more attention with the gradual improvement of hardware computing power. Until now, deep convolutional neural networks (CNNs) have been used in applications including SAR image classification, multiview SAR ATR, and polarimetric SAR image recognition [13]–[15]. The algorithm that integrates feature extraction and classifier demonstrates its powerful recognition performance.

However, data-driven recognition algorithms still rely heavily on the capacity of the data set [16]. In other words, the sufficient sample set assumption has been considered an objective hypothesis for the ATR model. The emergence of the small sample set seems to break this assumption. Therefore, utilizing a small sample set to train SAR ATR models has gradually become a topic of concern in recent years. There are usually two effective ways to solve the small sample set issue [17]. Data synthesis approaches can be used to overcome the problem of a small sample set in a straightforward manner [18], [19], and transfer learning has also shown potential for solving the small sample set problem [20], [21].

The rapid development of deep CNNs in ATR technology is inseparable from the support of a sufficient sample set. Meanwhile, it is easy to regard the small sample set problem as an accident of the sufficient sample set assumption. Interestingly, the two seemingly contradictory scenarios of sufficient

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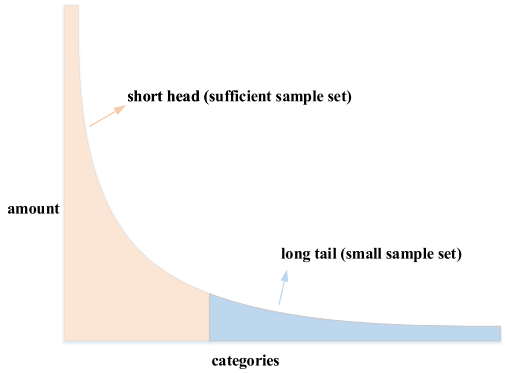


Fig. 1. Long tail data distribution schematic. The long tail data distribution is a random variable distribution, which is more common than the normal distribution.

sample set assumptions and small sample set problems that exist simultaneously for the training set of the current ATR model. Moreover, the data used to train the ATR model happen to follow the long-tail distribution shown in Fig. 1, where data for a few categories are abundant (majority categories) while data for others are scarce (minority categories) [22]. It shows that using small sample sets to describe the current problems in the training set used for the ATR model is not comprehensive.

The essence of the current problem facing the training data of the ATR model should be comprehended as an "imbalanced data problem". At this time, the problem that the minority categories bring to the ATR model is not just insufficient information. More importantly, the imbalanced data distribution forces the classification algorithms to be biased toward the majority categories [23]. As a result, the characteristics of the minority categories are not adequately learned. It is easy to feed the arbitrariness of the ATR model to misjudge the target sample as a majority category sample. However, some infrequent minority category samples may be extremely important recognition value for the ATR model.

Unfortunately, there have been few works dealing with imbalanced data for ATR technology in recent years. In particular, deep CNNs, as an important existing technical means of ATR, still cannot solve the problem of imbalanced data. Therefore, it is necessary to explore ATR technology that addresses this issue. Drawing on the current perspective of optical target recognition, the imbalanced data problem is usually approached from either the data level or algorithm level.

The data-level approaches usually deal with the imbalanced data problem by balancing the number of training samples. The existing data-level approaches can be roughly divided into two categories: 1) oversampling-based methods that balance the training set by synthesizing minority category target samples [24], and 2) undersampling-based methods that balance the training set by reducing the majority category target samples [25]. However, oversampling-based methods tend to increase the adverse correlations among the target samples [26]. On the other hand, undersampling-based methods solve the problem of imbalanced data while also losing a lot of useful discrimination information. Ultimately, it is difficult for data-level approaches to fundamentally solve the imbalanced data problem.

Instead of manipulating target sample sets, algorithm-level approaches adjust the cost of misrecognition by the algorithm [27]. Algorithm-level methods usually construct a cost matrix, which corresponds penalties to the recognition results in different situations. Huang *et al.* [28] proposed a label embedding method, which takes the misrecognition cost of interest into account. Castro *et al.* [29] utilized a single cost parameter to discriminate the importance of misrecognition cost. However, algorithm-level approaches often rely heavily on expert experience. Therefore, determining the punishment cost becomes a difficult and debatable issue for most ATR models.

This article proposes a cost-sensitive awareness-based automatic target recognition (CA-ATR) model to deal with imbalanced data. First, the proposed approach performs feature balanced reestablishment on the imbalanced training set in order to decrease its effect on the ATR model. Then, it embeds a cost matrix in the conventional neural network architecture so that each recognition result of the target sample gets a corresponding penalty. It avoids the arbitrariness of conventional CNN to distinguish minority category samples as majority category samples. At the same time, the minority category samples are also fully represented in the loss function. Moreover, the proposed method can autonomously learn the cost of each recognition result from the training set, thus avoiding the dependence on expert experience, making the CA-ATR model a more generalized approach.

The remaining section of this article is organized as follows: In Section II, the related work of the proposed approach is introduced; In Section III, the detailed model framework and implementation details of the proposed method are given; Section IV presents the experimental results of the proposed method, and Section V introduces conclusions of the proposed method.

II. Related Works

A. Current Data-Level Approaches

Although data-level approaches cannot fundamentally solve the imbalanced data problem, those methods are still an important means to alleviate the detriments of imbalanced data. Among them, oversampling and undersampling methods attempt to address imbalanced data, but also have many drawbacks. Adverse correlations among target samples make it easy for ATR models to overfit some unfavorable features. Moreover, losing a lot of useful discrimination information is equally fatal for ATR models. However, compared with the direct harm caused by a small sample set, the drawbacks of oversampling-based methods seem to be more acceptable, motivating its further development.

Among oversampling-based methods, the data augmentation method has been widely used in various fields of deep learning since its inception and has become a routine method [30]. Flip, rotate, and crop as important means of data augmentation have proven their effectiveness.

Moreover, Chawla *et al.* [31] proposed a method, called synthetic minority oversampling technique (SMOTE) to generate

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new target samples by linear interpolation between closely lying minority category samples. For each randomly selected neighbor sample x^*, a new sample x_{new} is constructed with the original sample x according to the following formulation:

$$x_{\text{new}} = x + \text{rand}(0, 1) * |x - x^*|. \tag{1}$$

However, the selection of neighbor sample x^* is a random process. When the selected neighbor samples are at the edge of the distribution, the algorithm easily blurs the classification boundary between different categories of samples. It improves the balance of the training set but increases the difficulty of the recognition algorithm for the ATR model. However, the emergence of improved algorithms, such as safe-level SMOTE [32] and local neighborhood SMOTE [33], gradually solved the problems of the original SMOTE algorithm. Unfortunately, there are few works that apply the SMOTE algorithm to ATR technology. Whether the SMOTE algorithm is effective for SAR target images is still an unexplored problem. At the same time, the combination of undersampling and SMOTE procedures to balance the training set has also been shown to produce better performance [34], [35].

Different from the SMOTE algorithm, generative adversarial nets (GANs) as an emerging data synthesis method have received widespread attention in recent years [18], [36]–[38]. Cao et.al proposed an SAR target sample synthesis method, called label directed generative adversarial nets (LDGAN), which is used to generate target samples of specified categories. Moreover, LDGAN has also shown the potential to use the generated samples for the expansion of the ATR model training sample set.

B. Current Algorithm-Level Approaches

Different from data-level approaches, algorithm-level approaches are more focused on processing the imbalanced data from the recognition cost. These approaches stem from a drawback of the existing ATR algorithm. The purpose of the ATR model is to minimize the expected risk of the recognition result to the target sample set. For example, given an input target sample x, the purpose of the ATR model is to minimize the expected risk $R_{\text{ex}}(y|x)$, where y is the label predicted by the ATR model. The formulation of expected risk can be expressed as

$$R_{\text{ex}}(y|x) = \sum_{y} q(y|x) \tag{2}$$

where $q(y|x)$ is the posterior probability over all categories for the target sample x. Since the posterior probability cannot be easily obtained, the ATR model usually obtains an empirical risk from the training sample set to replace it. Given the training set $T = \{x_i, y_i\}_M$, where M is the number of training sample set, the empirical risk $R_{\text{em}}(y|x)$ of ATR model can be formulated as follows:

$$\arg\min \mathrm{R}_{\text{ex}}(\mathrm{y}|\mathrm{x}) = \arg\min \mathrm{R}_{\text{em}}(\mathrm{y}|\mathrm{x}) = \frac{1}{M}\sum_{i=1}^{M} l(y_i, o_i) \tag{3}$$

where o_i is the actual output of the ATR model for a target sample x_i. $l(\bullet)$ is an optional loss function of the ATR model. However, these recognition methods all imply the following form of recognition cost:

$$C = \begin{cases} c_{p,k} = 0, & p = k \\ c_{p,k} = 1, & p \neq k \end{cases} \tag{4}$$

$c_{p,k}$ refers to the cost of recognizing the pth category sample as the kth category sample. In other words, different misrecognition results have the same recognition cost for the loss function.

It indicates that only when the training sample set is a balanced data set can the empirical risk $R_{\text{em}}(y|x)$ be close to the expected risk $R_{\text{ex}}(y|x)$. Otherwise, the ATR model often pays more attention to the contribution of majority categories to the loss function to minimize the empirical risk. Taking the mean squared error (MSE) function as an example, it is easy to know that the MSE function can be expressed as the following form:

$$l = \frac{1}{M}\sum_{i=1}^{M} \frac{1}{2}(y_i - o_i)^2. \tag{5}$$

The recognition cost of different types of target samples is the same. Because the minority category samples account for a small proportion of the training set, the minority category samples contribute little to minimizing the loss function. In the pursuit of minimizing empirical risk, ATR models often pay more attention to the fact that the majority categories of samples can be recognized as accurately as possible. It is easy for the ATR model to ignore the contribution of minority category samples, thereby reducing their recognition rate.

In this regard, Wang *et al.* [39] proposed the mean squared false error (MSFE) function to solve the problem of imbalanced data sets. The authors proposed the false-positive errors (FPE) and false-negative errors (FNE) inspired by the confusion matrix as follows:

$$\text{FPE} = \frac{1}{P}\sum_{i=1}^{P} \frac{1}{2}(y_i - o_i)^2 \tag{6}$$

$$\text{FNE} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{2}(y_i - o_i)^2 \tag{7}$$

where P and N represent the number of positive samples and negative samples, respectively. Then, the MSFE function is defined as follows:

$$\begin{aligned} \text{MSFE} &= \text{FPE}^2 + \text{FNE}^2 \\ &= \frac{1}{2}(\text{FPE} + \text{FNE})^2 + \frac{1}{2}(\text{FPE} - \text{FNE})^2. \end{aligned} \tag{8}$$

At this time, the contribution of the positive sample (majority categories) and the negative sample (minority categories) to the loss function can be kept consistent. However, as an algorithm for processing imbalanced data, the MSFE function is more focused on solving binary classification problems. It is regrettable that almost all ATR tasks are unlikely to be binary classification tasks. Therefore, such a loss function is difficult to generalize to ATR tasks. At the same time, it is worth noting that this proves that imposing different penalties on different recognition results can solve the problem of imbalanced data sets.

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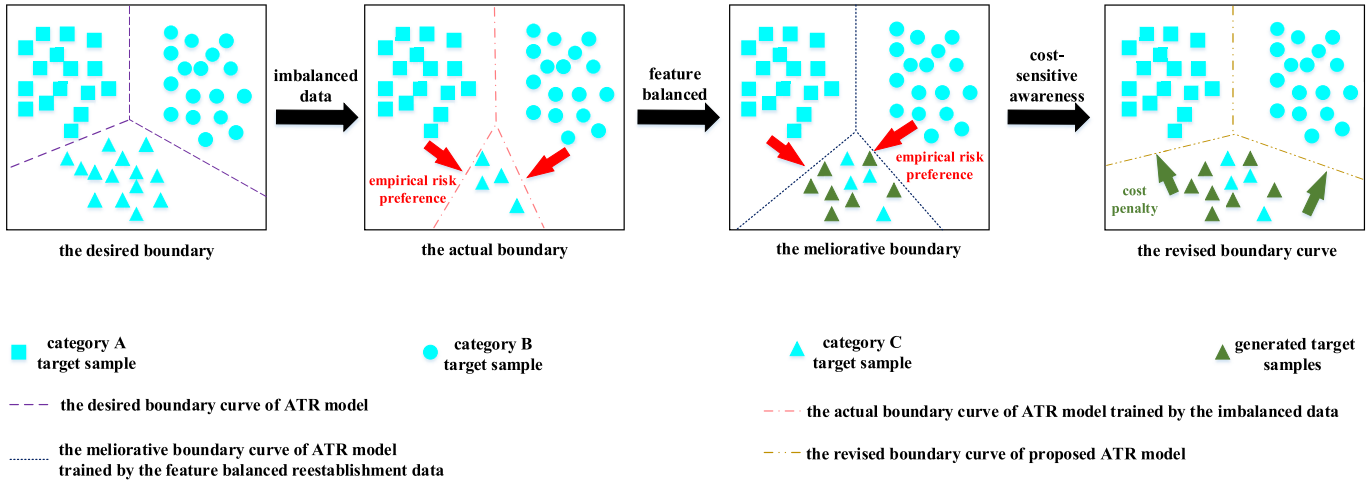


Fig. 2. Detailed motivations about the proposed approach. The purple dotted line represents the desired boundary curve of ATR model. The red dotted line represents the actual boundary curve of ATR model trained by the imbalanced data. The black dotted line represents the meliorative boundary curve of ATR model trained by the feature balanced reestablishment data. The brown solid line represents the revised boundary curve of proposed ATR model.

Similarly, according to the above ideas to solve the problem, focal loss has gradually been applied to deal with imbalanced data classification [40]. Focal loss is a modification based on the cross-entropy loss function. Taking the binary classification cross-entropy loss as an example, the loss function of focal loss is as follows:

$$L_{fl} = \begin{cases} -(1-y_i)^{\gamma} \log y_i, & \mathrm{y_i} = 1 \\ -y_i^{\gamma} \log(1-y_i), & \mathrm{y_i} = 0. \end{cases} \tag{9}$$

Among them, the purpose of γ is to reduce the loss of majority category samples so that the recognition model pays more attention to minority category samples. However, it is not enough to focus only on the contribution of the minority category samples to the loss function. It is better to adjust the misrecognition cost of different target samples from an overall perspective.

III. Proposed Method

A. Motivations for the Proposed Method

The general ATR models usually decline seriously in recognition performance when the imbalance ratio (IR) of training data is high. Among them, the ATR model represented by the deep learning algorithm still cannot avoid the problems caused by the imbalance of the data set. Imbalanced training sets usually cause two problems for ATR models.

First, the number of minority category samples is smaller than that of the majority category samples, which is not just a problem of the imbalanced training set. In particular, when the IR of the training set is large, the recognition performance of minority category samples is likely to suffer from the small sample set problem. At this time, the number of minority category samples is small, which means that the effective discriminant information of the ATR model for minority category samples is also drastically reduced. The lack of effective discrimination information directly leads to the decline of the ATR model's recognition performance for minority category samples.

Second, the purpose of the ATR model is to minimize the expected risk of the recognition result to the target sample set, which means that the ATR model is more focused on the overall risk aversion of the training set. The overall risk aversion of the training set shows that the goal of the ATR model is to minimize the sum of the expected risk by all training samples. In fact, the ATR model usually obtains an empirical risk from the training sample set to replace the expected risk. Therefore, if the number of a certain category target sample in the training set is small, it is difficult to greatly affect the sum of the empirical risk. It leads the ATR model to pay more attention to minimize the sum of empirical risks for majority categories samples. The ATR model wants to increase the recognition performance of majority category samples as much as possible, which results in the preference of the ATR model for majority category samples. The preference further exacerbates the decline in the recognition performance of minority category samples. In other words, the majority category samples invade the area that belongs to the minority category samples.

The proposed CA-ATR method solves both problems by addressing them through two approaches, as shown in Fig. 2. The insufficient number of training samples easily leads to the lack of effective discrimination information for the ATR model and the features of minority category samples are not fully learned by the ATR model. However, the minority category samples in the imbalanced training set are often difficult to obtain in reality. Therefore, the proposed method needs to be able to improve the dilemma of insufficient minority category samples and should be able to balance the features of the training set in some way.

Moreover, the empirical risk preference of the ATR model for majority category samples has not been corrected. The empirical risk preference should be corrected or devalued so that minority category samples are equally valued in the recognition process. Therefore, the proposed method penalizes the empirical risk preference of the ATR model so that each category of sample is treated equally. We believe that a

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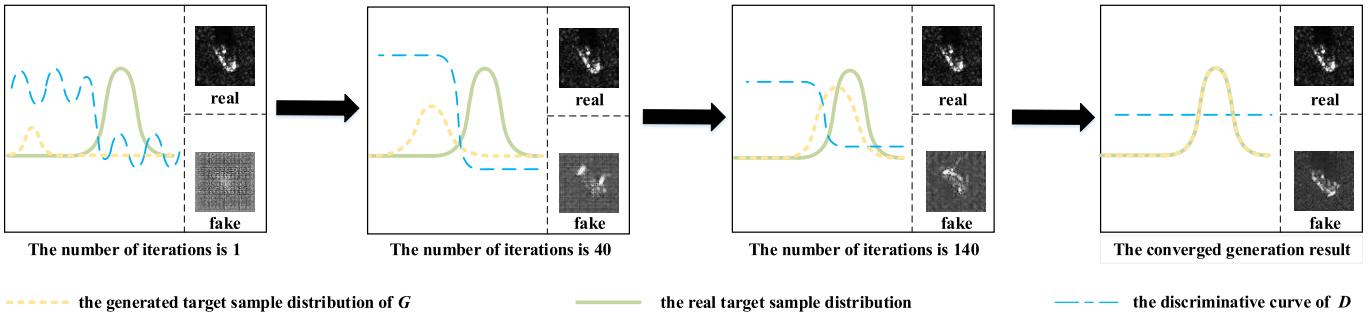


Fig. 3. Schematic of GANs generating samples. The yellow dotted line represents the generated target sample distribution of G. The green solid line represents the real target sample distribution. The blue dotted line represents the discriminative distribution of D.

cost-sensitive ATR model can effectively solve the imbalanced data problem on the premise that the imbalanced features of the training set are improved.

B. Feature Balanced Reestablishment

The significance of feature balanced reestablishment is to improve the feature imbalance of the training sample set. At the same time, the adverse correlation among the target samples causes the ATR model to overfit some unnecessary features. Therefore, the proposed feature balanced reestablishment utilizes two data synthesis methods to decrease the adverse correlation among the target samples. First, the oversampling of minority category samples is achieved through flip and rotation. Second, GANs are utilized to generate minority category samples. By combining these two data generation methods, the feature balanced reestablishment of the imbalanced data set is realized.

GANs are a new type of data generation algorithm that has received widespread attention in recent years. The architecture of GANs consists of a deconvolutional and a convolutional network. The deconvolutional network, called generator G, is trained to generate samples from data distribution $P_r(x)$, by changing the noise z from noise distribution $P_z(z)$ to $P_r(x)$, where $x = G(z)$. The convolutional network, called discriminator D, is trained to determine whether the sample generated by the generator is a real or fake sample during the process of training.

The schematic of GANs generating samples is shown in Fig. 3. The GANs use noise as the initial result of the network. When GANs start to iterate, there is a significant difference between the generated target sample distribution and the data distribution. At this time, the recognition confidence of the discriminative curve is also unstable for the real sample and the generated sample. As the number of iterations increases, the generated target sample distribution gradually approaches the real data distribution. At the same time, the recognition confidence of the discriminative curve also tends to be stable. After many iterations, the gradient of D has guided $G(z)$ to flow to regions that are more likely to be classified as real target samples. The recognition confidence of the discriminative curve shows a gradual decline. When GANs converge, the discriminant curve can no longer distinguish whether the sample is real. The discriminant accuracy rate of the discriminant curve becomes 0.5.

It is difficult to generate samples using GANs, especially SAR samples, due to the unreasonable selection of distance measurements for GAN. Therefore, we have utilized another kind of distance measurement called Wasserstein distance to replace the original distance measurement. The formulation expression of Wasserstein distance is as follows:

$$W(P_r(x), P_g(y)) = \inf_{\gamma \in \prod(P_r(x), P_g(y))} E_{(x,y)\sim\gamma}[||x - y||] \quad (10)$$

where $\prod(P_r(x), P_g(y))$ is the set of all the joint distributions $\gamma(x, y)$, and its marginal distributions are $P_r(x)$ and $P_g(y)$. P_g represents the generated target sample distribution.

At the same time, taking into account the need to generate the required target samples by category, the loss function form used by GANs is as follows:

$$\begin{aligned}\text{Loss} = {} & E_{x_g\sim P_g}[D(x_g|y)] - E_{x_r\sim P_r}[D(x_r|y)] \\ & + \lambda E_{x\sim P_x}[(||\nabla_x D(x|y)||_2 - 1)^2] \quad (11)\end{aligned}$$

where y represents the label information, which is the key to being able to generate target samples by category. λ is expressed as a regularization coefficient. Besides, sample a random number $\xi \sim U[0, 1]$, the expression of x of (11) is as follows:

$$x = \xi x_r + (1 - \xi)x_g. \quad (12)$$

Moreover, it cannot be ignored that GANs also suffer from the impact of imbalanced data sets. Therefore, the proposed method must account for this problem in the implementation process.

C. Loss Function About the Cost-Sensitive Awareness

Since CNN algorithms have been shown to perform the best in the ATR field, this article focuses on exploring the construction of cost-sensitive awareness of the CNN ATR model. There is no doubt that the CNN model is also affected by empirical risk preference since the model assumes that the cost corresponding to the recognition result of each target sample is the same. However, the recognition result of the CNN ATR model prefers the majority category samples when provided imbalanced training data. Therefore, the same cost of recognition results in unequal treatment.

Generally, different cost thresholds can be used to punish different recognition results, so that the risk preference of the

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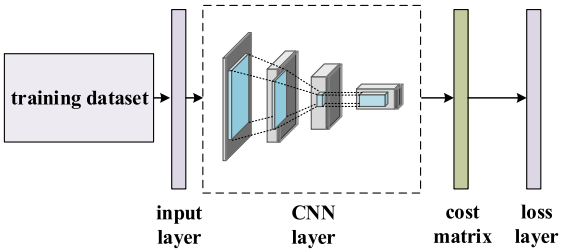


Fig. 4. Schematic of cost-sensitive awareness CNN.

ATR model can be corrected. The penalty is usually applied after the ATR model obtains the recognition result. However, the step is based on an incorrect assumption. For example, in the moving and stationary target acquisition and recognition (MSTAR) task, the ATR model recognizes 10×10 results in total, but only 10 are obtained. Therefore, it is not enough to only punish the 10 final results. Moreover, the penalty threshold often depends on expert experience and lacks generalization. Thus, the proposed method embeds the penalty for 100 recognized results in the CNN model. At the same time, the architecture enables the ATR model to learn independently of the training set to differentiate penalties for 100 recognized results. The schematic of cost-sensitive CNN is as shown in Fig. 4.

The training details of cost-sensitive CNN is introduced in Section III-D. This section focuses on the derivation of the loss function of the proposed CNN architecture. To make the proposed method generalizable, we choose the most widely used cross entropy (CE) function in the CNN architecture to derive the new loss function. The CE loss function is as follows:

$$H(p,q) = -\sum_{x} p(x) \log q(x) \tag{13}$$

where $p(x)$ is the expected output of the target sample. $q(x)$ is the actual output of the target sample, and its detailed form is as follows:

$$q(x) = \frac{\exp(y_i)}{\sum_{k=1}^{n} \exp(y_k)} \tag{14}$$

where y_i is the result of the ith logical unit corresponding to a target sample. y_k is the result of all the logical units corresponding to a target sample. The result of $q(x)$ is the output of the CNN layer. As shown in Fig. 4, the result of $q(x)$ is no longer directly input to the loss layer but pass a cost matrix. The cost matrix is a matrix of order $C * C$, where C is the number of categories contained in the training sample set. C_{pi} refers to the cost of recognizing the pth category sample as the ith category sample. Therefore, the proposed actual output of the target sample $q^*(x)$ is as follows:

$$q^*(x) = \frac{c_{\text{pi}} \exp(y_i)}{\sum_{k=1}^{n} c_{pk} \exp(y_k)}. \tag{15}$$

Substituting (15) into (13), the cross-entropy loss function becomes the following form:

$$H^*(p,q) = -\sum_{x} p(x) \log \frac{c_{\text{pi}} \exp(y_i)}{\sum_{k=1}^{n} c_{pk} \exp(y_k)}. \tag{16}$$

Moreover, each value of the cost matrix has a practical meaning, so several mathematical constraints need to be added to the cost matrix. The value of C_{pi} should be a positive number because C_{pi} is a recognition cost. At the same time, C_{ii} is a cost of correct recognition and its value should be less than the misrecognition cost C_{pi} $(p \neq i)$. Therefore, the proposed loss function of cost-sensitive CNN is as follows:

$$H^*(p,q) = -\sum_{x} p(x) \log \frac{c_{\text{pi}} \exp(y_i)}{\sum_{k=1}^{n} c_{pk} \exp(y_k)}$$
$$\text{s. t.} \begin{cases} c_{\text{pi}} \geq 0 \\ c_{\text{ii}} \leq c_{\text{pi}} (p \neq i). \end{cases} \tag{17}$$

D. Implementation of CA-ATR Model

CA-ATR strives to implement the following improvements: 1) enable the ATR model to learn the features of minority category samples more fully, and 2) correct the empirical risk preference of the ATR model for the majority category target samples. Based on the previously discussed problems that the proposed improvements address, the complete framework of CA-ATR is shown in Fig. 5. However, some details still need to be taken into consideration in order to implement it.

In general, it cannot be ignored that GANs also suffer from the impact of imbalanced data sets. The generated results of GANs depend on the feature balance of the training sets. Fortunately, relevant literature has proven the ability of GANs to supplement target sample information under the condition of a small sample set [19]. It also shows that it is unnecessary to worry that the loss of discrimination information caused by undersampling affects the performance of GANs. Therefore, the proposed method implements the undersampling of majority category samples to balance the features of the training set. The undersampled majority category samples and the minority category samples constitute the training sample set of GANs. The proposed method utilizes this training sample set for target sample generation.

Moreover, it is not natural to use the samples generated by GANs to directly improve the performance of the ATR model. It is difficult for GANs to avoid overfitting when there are few samples. Therefore, we need to set some prerequisites to determine whether the generated samples can be used to improve the performance of the ATR model, which we accomplished by filtering the generated samples and only using those to train the ATR model. For instances of insufficient discrimination information, an SVM is undoubtedly a better choice than CNN. At the same time, it has been our experience that the filtered generated samples have advantages in improving the performance of the ATR model. The subsequent experimental results readily prove the effectiveness of the proposed method under all experimental conditions in this article.

The imbalance of the training sample set is improved through feature balanced reestablishment. However, the empirical risk preference of the ATR model for majority category samples has not been corrected. The purpose of cost awareness construction is to prevent the ATR model from showing preference to majority category samples. The cost awareness construction enables the ATR model to learn independently of

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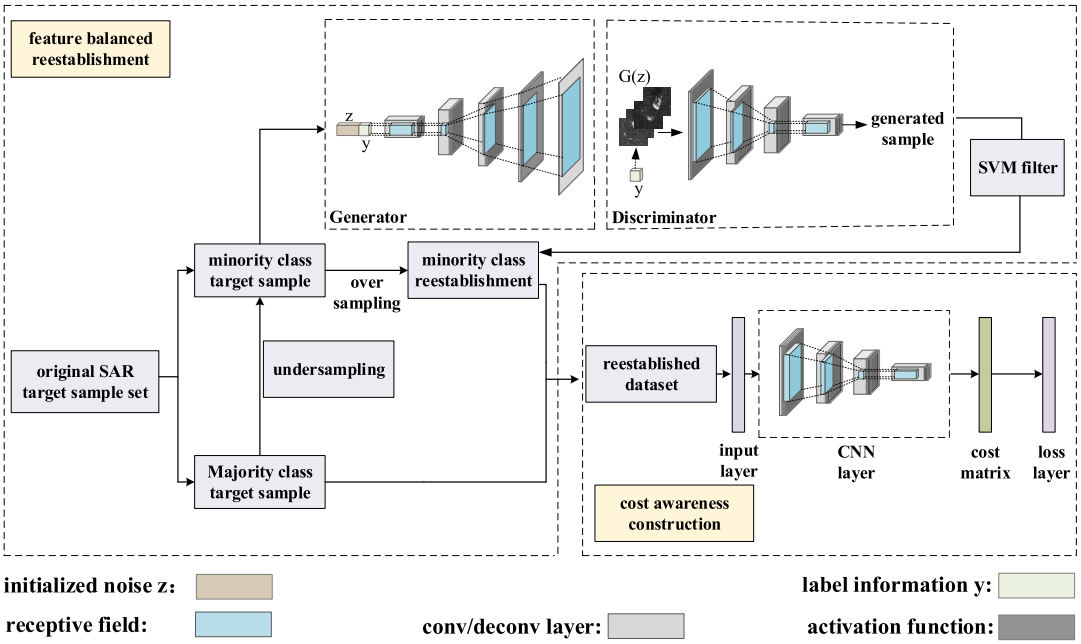


Fig. 5. Complete framework of the proposed method. The significance of feature balanced reestablishment is to improve the feature imbalance of the training sample set. The purpose of cost awareness construction is to make the ATR model no longer exist the preference for majority categories samples.

the training set to differentiate penalties for $C * C$ recognition results. The score-level penalties directly perturb the recognition results of the ATR model and the different perturbations enable the ATR model to pay more attention to the minority samples, thereby avoiding the empirical risk preference of the ATR model for majority category samples.

However, it also follows that cost-sensitive CNN cannot achieve convergence according to the conventional CNN training method. Therefore, this section specifically explains the unique training method of cost-sensitive CNN. As shown in Fig. 5, the cost matrix is embedded in the CNN. However, the cost matrix is different from the fully connected layer of CNN, and it is also different from the convolutional layer. The optimization goal of cost-sensitive CNN is to minimize the empirical risk from the training sample set under the condition of such perturbation. In other words, the empirical risk is the result of the joint action of the CNN parameters θ and the cost matrix parameter c. These two parameters affect each other, and both cause changes in the recognition accuracy of the ATR model.

Therefore, the proposed iterative optimization method adopts a strategy that fixes a type of parameter to optimize another type of parameter to minimize the empirical risk of the ATR model. At the same time, the optimization difficulty of the cost matrix parameter c is much lower than the optimization difficulty of the CNN parameters θ. It is also the reason why that the cost matrix is optimized only once in an epoch. The detailed training method of cost-sensitive CNN is shown in algorithm 1.

IV. Experimental Results

The imbalanced data problem is present in nearly all real-world objects and image data sets. The only difference is the degree of imbalance of the data set, which has different impacts on the performance of the ATR model. The problem is not caused by any defective data collection method, but the naturally different frequency patterns of different target categories in real life. For example, there are obvious differences in the natural frequency patterns of the appearance of different military targets in real life. From the perspective of the imbalanced data problem, the current problem of the ATR model can be divided into two categories: 1) the impacts of different proportions of minority categories in the training sample set on the ATR model, and 2) the impact of the IR of the training sample set on the ATR model. The IR refers to the ratio between the number of samples in the majority categories and the number of samples in the minority categories.

Experiments were carried out to evaluate the recognition performance of the proposed method for imbalanced data processing. The experiments were performed on the MSTAR database, which is a commonly used data set for SAR ATR experiments and contains 0.3 m $\times$ 0.3 m resolution's SAR target samples obtained from X-band, HH-polarized, and bunching mode sensors [41]. Ten categories of SAR target samples were selected in the process of these experiments. The samples used in this experimental process were slices of MSTAR from ten SAR categories. All the slices were 64 $\times$ 64 RAW data as shown in Fig. 6. The training samples consisted of SAR target samples with a pitch angle of 17°. Additionally, the experiment also used SAR target samples with a pitch angle of 15° as the test samples. The number of target samples in the MSTAR data set is shown in Table I.

A. Damage Caused by Imbalanced Data

The imbalanced data problem causes the degradation of the ATR model. At the same time, the small sample set problem

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Algorithm 1 Unique Iterative Optimization for Cost-Sensitive Awareness CNN Parameters (θ, c)

Require: Training Sample Set (X, Y), Learning Rate r_θ for θ, Learning Rate r_c for c, the Attenuation Rate a_θ of Learning Rate r_θ, the Attenuation Rate a_c of Learning Rate r_c, Max Epochs $\max_{\text{ep}}$, the Batch Size m_{bat}, the Total Number of Training Samples M

Require: The Initial Cost-Sensitive CNN Parameters θ, the Initial Cost Matrix Parameters c. All the Initialization Parameters of the Cost Matrix Must Meet the Constraints.

while $H^*(p, q)$ has not converged **do**
  for $e \in [1, \max_{\text{ep}}]$ **do**
    freeze θ while **unfreeze** c
    $Grad_c \leftarrow backward - pass(X, Y)$
    $c^* \leftarrow gradient - descent(Grad_c, c, r_c, a_c)$
    $c \leftarrow c^*$
    for $b \in [1, \left\lfloor \frac{M}{m_{mat}} \right\rfloor]$ **do**
      freeze c while **unfreeze** θ
      $out_b \leftarrow forward - pass(x_b, y_b, \theta)$
      $Grad_b \leftarrow backward - pass(out_b, x_b, y_b, \theta, c)$
      $\theta^* \leftarrow Adam(Grad_b, \theta, r_\theta, a_\theta)$
      $\theta \leftarrow \theta^*$
    end for
  end for
end while
return parameters (θ, c)



Fig. 6. Ten categories of SAR target samples. The order from left to right of each row of images is as follows: 2S1, BMP2, BRDM2, BTR60, BTR70, D7, T62, T72, ZIL131, and ZSU23/4. Row1: the optical images corresponding to ten categories of target samples. Row2: the SAR target samples of ten categories of target samples.

TABLE I
SUMMARY OF THE MSTAR DATA SET

Target Type	Serial No.	# Images (17°)	# Images (15°)
2S1	b01	299	274
BMP2	9563	233	195
BRDM2	E-71	298	274
BTR60	k10yt7532	256	195
BTR70	c71	233	196
D7	92v13015	299	274
T62	A51	299	273
T72	sn-132	232	196
ZIL131	E12	299	274
ZSU23/4	d08	299	274

also decreases the recognition performance of the ATR model. However, the imbalanced data problem brings about the small sample set problem. Therefore, how do we determine that the degradation of the recognition performance is not only caused by the small sample set problem? The first experiment removed the impact of the small sample set problem on the ATR model and showed the ATR model's empirical risk preference for majority category samples.

The first experiment built two training sample sets: the small sample data set and the imbalanced data set. The small sample data set consisted of 40 samples randomly selected from each type of target sample to form a new training sample set. In the imbalanced data set, the experiment selected three categories of target samples, 2S1, BTR60, and T62, from the small sample data set to replace the corresponding target samples in the original data set. The experiment used two different training sets to train the CNN model of the same architecture and obtained the recognition results shown in Fig. 7.

As shown in Fig. 7, the recognition rate of the majority category target samples showed a significant drop under the influence of the small sample set problem. At the same time, the experiment also reflected a phenomenon contrary to common sense. The target samples 2S1, BTR60, and T62 contained in the two different data sets were the same. Moreover, the imbalanced data set contained more target samples of other categories than that of the small sample data

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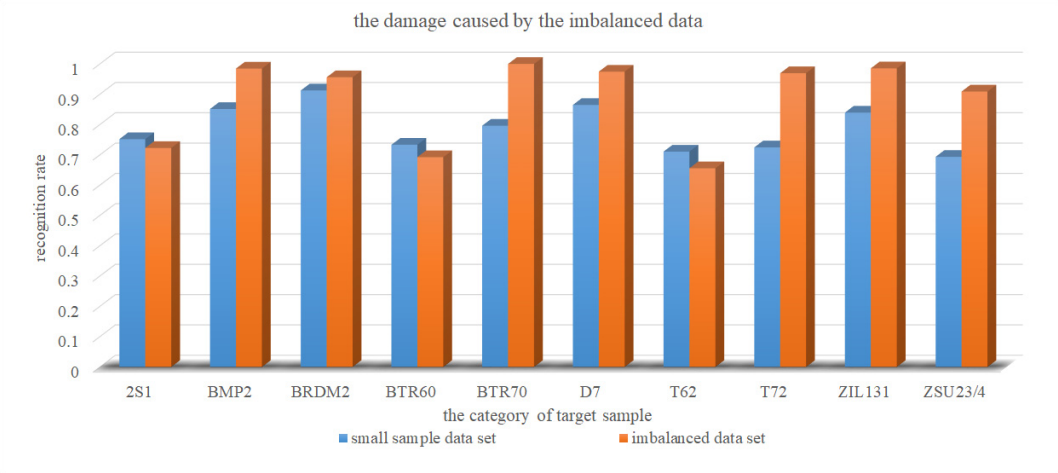


Fig. 7. Damage caused by the imbalanced data. The imbalanced training set brings two problems to the ATR model: insufficient discrimination information and the empirical risk preference for majority categories samples.

set, which means that the imbalanced data set contained more discrimination information for the target sample. However, the recognition performance of the ATR model with more discrimination information was declined when faced with these three target sample categories. This phenomenon was caused by the empirical risk preference of the ATR model for majority category samples. The number of target samples from the other seven categories was far greater than the number of target samples from the three selected categories. To minimize the empirical risk of the ATR model, it was easy to ignore the recognition accuracy of these three categories of minority target samples. Therefore, the imbalanced training set caused two problems for the ATR model: insufficient discrimination information and the empirical risk preference for majority category samples.

B. Effectiveness of CA-ATR Components

According to the recognition results of the previous experiment, insufficient discrimination information and empirical risk preference affect the recognition performance of the ATR model. To address these effects, the proposed method utilizes feature balanced reestablishment and cost awareness construction to solve the two problems separately. Therefore, the second experiments evaluate the effectiveness of these two proposed components.

Feature balanced reestablishment used two data synthesis methods to balance the feature of the imbalanced training set. In addition, the oversampling and undersampling methods are also ways to solve the feature imbalance problem from the data-level perspective. The first experiment compared the recognition performance of the above three feature balancing methods to verify the effectiveness of the feature balanced reestablishment. The experiment also utilized the imbalanced data set used in the previous experiment. At the same time, this experiment reconstructed three new data sets based on the imbalanced data set: the oversampling training data set, the undersampling training data set, and the feature balanced reestablishment training data set. The oversampling training data set used the flip and rotation methods to increase the number of minority category target samples to be consistent with the majority category target samples. The undersampling training data set selected the same number of target samples from the majority category target samples as the minority categories target samples to balance the features of the training set. The experiment also implemented the data set reestablished in Fig. 5 to construct the feature balanced reestablishment training data set. The data set consisted of two data augmentation methods, which effectively avoided the adverse correlation among the target samples. The target sample numbers of the four training sets are shown in Table II. This experiment used these four training sets to train the CNN with the same architecture to observe their effects on the ATR model.

TABLE II
TARGET SAMPLE NUMBERS OF THE FOUR TRAINING SETS

category	imbalanced	oversampling	undersampling	feature reestablishment
2S1	40	299	40	299
BMP2	233	233	40	233
BRDM2	298	298	40	298
BTR60	40	256	40	256
BTR70	233	233	40	233
D7	299	299	40	299
T62	40	299	40	299
T72	232	232	40	232
ZIL131	299	299	40	299
ZSU23/4	299	299	40	299

However, it is difficult to generate samples using GANs, especially for SAR target samples. Currently, it is still uncertain whether the GAN used in this article can generate effective samples according to the steps in Fig. 5. Therefore, the experiments used the imbalanced data set in Table II to generate samples according to the steps in Fig. 5. The generated results of the proposed model are shown in Fig. 8. According to the results in Fig. 8, the proposed model was able to generate high-quality samples under this experimental condition.

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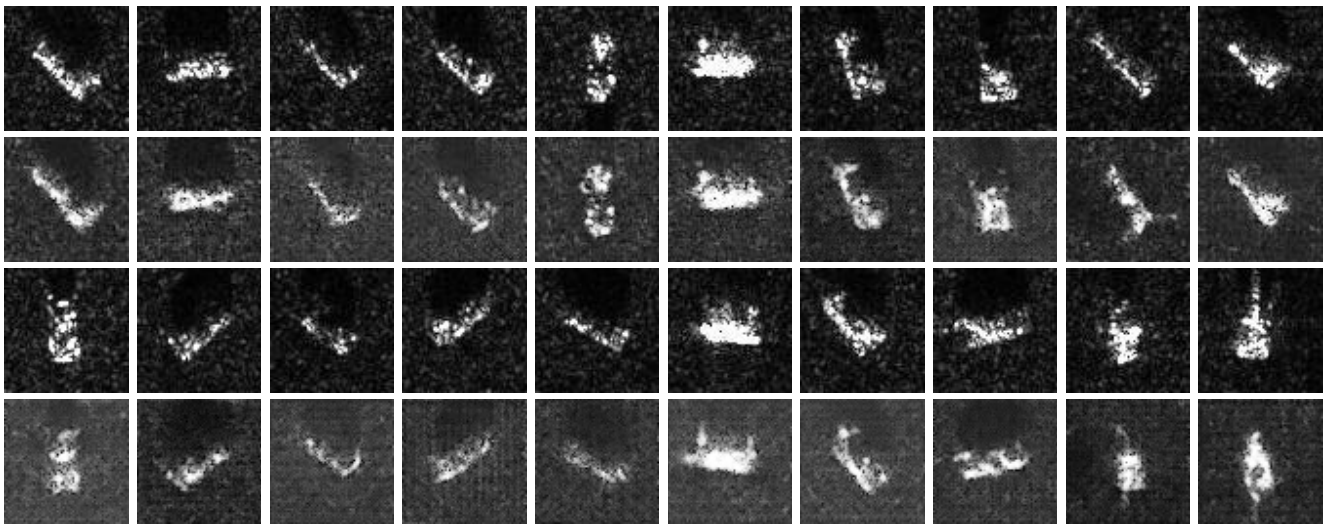


Fig. 8. Ten categories of SAR target samples. The order from left to right of each row of images is as follows: 2S1, BMP2, BRDM2, BTR60, BTR70, D7, T62, T72, ZIL131, and ZSU23/4. Row1: the real target samples of ten categories SAR target samples. Row2: target samples generated by the proposed model for ten categories of SAR target samples. Row3: the real target samples with another azimuth angle. Row4: the corresponding generated target samples with another azimuth angle.

Moreover, the experiments also used the four training sets, as shown in Table II, to train the CNN with the same architecture to record the recognition performance. The recognition performance of the four ATR models is shown in Fig. 9. As shown in Fig. 9, the three feature balanced methods improved the recognition rate of minority category target samples to varying degrees. Among them, the training set modified with feature balanced reestablishment increased the recognition rate of minority category target samples by 10.5% (2S1), 14.4% (BTR60), and 11.3% (T62). At the same time, this method improved the recognition rate of minority category target samples the most. Moreover, the experimental results also showed that different feature-balanced methods have different effects on the overall recognition rate of the ATR model. Even worse, the undersampling training set increased the recognition rate of minority category target samples while also causing the overall recognition rate of the ATR model to decrease. The feature-balanced reestablishment training set and oversampling training set increased the baseline recognition rate of the ATR model by 2.2% and 1.3%, respectively. Therefore, from the recognition rate of minority category target samples or the overall recognition rate of the ATR model, the feature balanced reestablishment was an effective method.

However, the experiment only proved the ability of GANs to supplement discriminative information when the IR was not more than 7.5 (the number of target samples per category was not less than 40). In order to ensure that subsequent experiments can be verified under broader conditions, we also separately verified the ability of GANs to supplement discriminative information with fewer samples. The experiment randomly selected three batches of sample sets from the training sample set again, the number of which were 360 (13.1% of the overall training set), 320 (11.6% of the overall training set), and 280 (10.1% of the overall training set). The proposed GANs were executed as shown in Fig. 5 under the conditions of four different numbers of samples, and four different batches of generated samples were obtained, each with 300 samples. When the information of the training sample enabled the recognition model to learn thoroughly, the subtle change in the recognition rate did not reflect that this change of the experiment was caused by the difference of target samples. Thus, the premise of this recognition experiment was based on the condition that the recognition model was seriously underfitting. The experiment randomly selected 100 samples from a real training target sample set. Next, the experiment used 50 samples as a batch and added the target samples generated under different sample numbers to the training samples in batches to retrain the model. Then, we recorded the recognition rate results under different experimental conditions. Finally, we obtained the experimental results shown in Fig. 10.

As shown in Fig. 10, when the proportion of training samples was not less than 10.1%, the proposed method still effectively improved the performance of the ATR model. In general, the proposed approach significantly improved the performance of the ATR model under different numbers of training samples. When the number of training samples was not less than 10.1%, the proposed method improved the original recognition rate of the ATR model by at least 13.95% under current experimental conditions. It showed that the generation model for ATR can provide more discriminative information under different sample numbers.

The feature-balanced reestablishment improved the feature imbalance of the training set. However, the ATR model trained by the imbalanced data set still exhibited an empirical risk preference for majority category samples. The second experiment verified the effectiveness of the cost awareness construct by using the ATR model trained by cost-sensitive CNN. The second experiment continued to use the four training sets shown in Table II. Moreover, all ATR models

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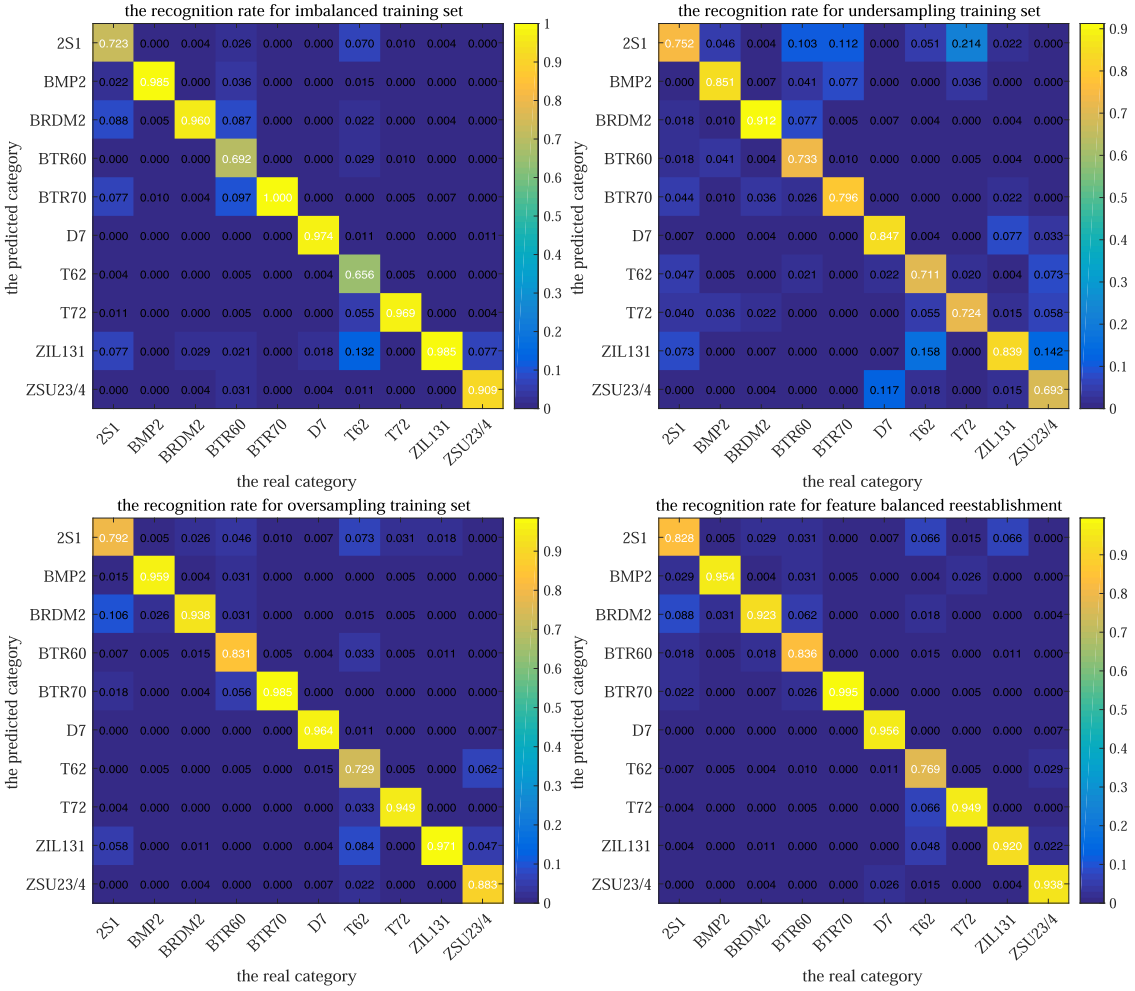


Fig. 9. Impact of four different training sets on the recognition performance of ATR model. The overall recognition rate of the four ATR models is: 88.2% (the overall recognition rate for imbalanced training set), 78.7% (the overall recognition rate for undersampling training set), 89.5% (the overall recognition rate for oversampling training set), and 90.4% (the overall recognition rate for feature balanced reestablishment training set).

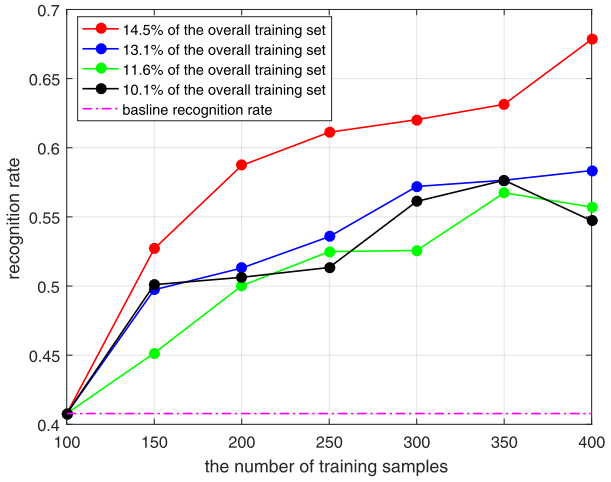


Fig. 10. When different numbers of samples are used to generate samples, the comparison result of recognition performance displayed by the ATR model is shown in the figure. Baseline recognition rate refers to the highest recognition rate that the ATR model can achieve when the number of training samples is 100.

of the second experiment were trained by the cost-sensitive CNN. The recognition performance of the cost-sensitive CNN trained by the four different training sets is shown in Fig. 11.

As shown in Fig. 11, the CNN ATR model with cost-sensitive awareness showed better recognition performance for the four training sets. The overall recognition rates of the four ATR models in the second experiment were all higher than the corresponding ATR models in the first experiment. Among them, the CNN ATR model with cost-sensitive awareness increased the baseline recognition rate by at least 1.9%. It verified the effectiveness of the cost awareness construction of the proposed method.

At the same time, such experimental results are inseparable from the cost-sensitive awareness of the CNN ATR model. Subsequently, the second experiment also showed the cost-sensitive awareness that the CNN ATR model had learned from the training set. Taking the feature balanced reestablishment training data set as an example, the cost of various recognition results learned by the ATR model is shown in Table III.

In Table III, the costs of various recognition results were all learned by CNN from the training set. The cost-sensitive awareness of CNN can be understood from three aspects. 1) The cost of correctly recognizing that the target sample is not zero, but this recognition cost was close to zero. 2) The cost of misrecognizing the target sample should not be the same. 3) When the number of some target samples is big

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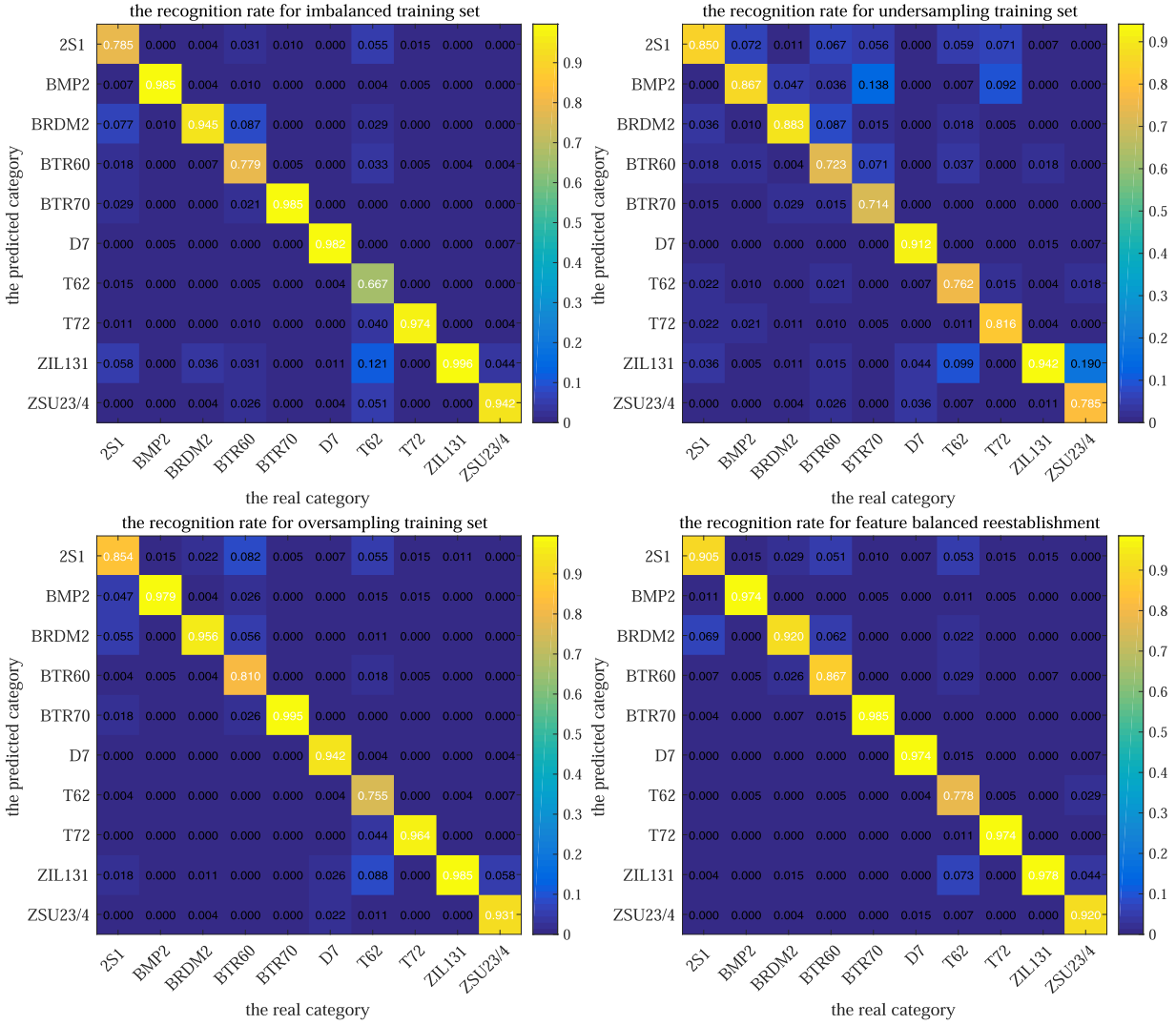


Fig. 11. Recognition performance of the cost-sensitive awareness CNN trained by the four different training sets. The overall recognition rate of the four ATR models is: 90.1%(the overall recognition rate for imbalanced training set), 83.1% (the overall recognition rate for undersampling training set), 91.4% (the overall recognition rate for oversampling training set), and 92.4% (the overall recognition rate for feature balanced reestablishment training set).

TABLE III

COST OF VARIOUS RECOGNITION RESULTS LEARNED BY THE ATR MODEL TRAINED BY THE FEATURE BALANCED REESTABLISHMENT TRAINING DATA SET

cost: predicted category / real category	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZSU23/4
2S1	4.5784e-04	5.5991	2.4004	2.3888	5.5939	2.3901	2.3826	5.3423	2.4013	2.3971
BMP2	2.3869	6.8224e-04	2.3936	2.3881	5.5884	2.4001	2.3992	5.3502	2.3934	2.3955
BRDM2	2.3788	5.5977	5.1869e-04	2.3772	5.5852	2.4015	2.4006	5.3559	2.3907	2.3944
BTR60	2.3813	5.6007	2.405	7.2115e-05	5.5893	2.3965	2.3983	5.3505	2.4014	2.3969
BTR70	2.3882	5.5791	2.3996	2.385	6.9844e-04	2.4032	2.3944	5.3591	2.4033	2.3966
D7	2.416	5.5475	2.3731	2.4485	5.5756	1.0392e-04	2.3942	5.3659	2.3928	2.396
T62	2.4011	5.5867	2.3932	2.3995	5.587	2.3887	1.5852e-04	5.3497	2.4014	2.3886
T72	2.3919	5.5875	2.3906	2.3946	5.5875	2.403	2.4007	4.9684e-04	2.3929	2.392
ZIL131	2.4039	5.5733	2.3967	2.4071	5.573	2.3943	2.3953	5.5326	2.4439e-05	2.3956
ZSU23/4	2.4091	5.5482	2.3812	2.4435	5.5647	2.3909	2.4057	5.3728	2.3886	7.3971e-04

and they are similar to other minority category target samples, the ATR model should increase the cost of misrecognition of such target samples instead of treating them equally. For example, as shown in Fig. 6, 2S1, BTR60, and T62 are similar to BMP2, BTR70, and T72, respectively, in morphological structure, which is easy to misrecognize. Therefore, when

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2S1, BTR60, and T62 are minority category target samples, the misrecognition cost of the corresponding majority category target samples (BMP2, BTR70, and T72) should increase significantly. It means that the ATR model was very cautious in recognizing the arbitrary target sample as these three categories of target samples. It is the reason that the cost-sensitive awareness of the proposed model can avoid the empirical risk preference of the ATR model.

Moreover, we also noticed an interesting experimental phenomenon. The purpose of the ATR model is to minimize the expected risk of the recognition result to the target sample set, which means that the ATR model is more focused on the overall risk aversion of the training set. Therefore, we may not determine the cost of a certain recognition result from a single recognition result. For example, as shown in Fig. 9, according to the confusion matrix results of the recognition rate for the imbalanced training set, 2S1 was more easily confused with BRDM2, BTR70, and ZIL131. The proposed model did not separately increase the cost of these three misrecognition results. However, the proposed method still significantly decreased the misrecognition rate of the ATR model. It shows that cost-sensitive awareness is an overall consideration of the ATR model for the current training sample set. The ATR model does not need to deliberately keep the recognition cost of minority target samples to the lowest. Otherwise, it may cause the overall recognition rate of the ATR model to drop. The experimental results also demonstrated that it is better to adjust the misrecognition cost of different target samples from an overall perspective.

C. Recognition Performance of CA-ATR Under Different Imbalanced Conditions

When training an ATR model with an imbalanced training set, there are usually two dilemmas: 1) the proportion of minority category target samples in the training set affects the recognition performance of the ATR model, and 2) the IR of minority category target samples also affects the recognition rate of the ATR model. Therefore, we evaluated the recognition performance of the proposed method from these two aspects.

The first experiment evaluated the impact of different proportions of minority category target samples in the training set on the recognition performance of CA-ATR under the premise of a fixed imbalance rate. The experiment fixed the IR in the training set to 7.5 and fixed the number of minority category target samples to 40. Therefore, all training data sets in the experiment were constructed according to this requirement. At the same time, the proportion of minority category target samples were increased from 10% to 90%, and the results of oversampling and undersampling were added in the comparison of experimental results. Moreover, an imbalanced training set damages the ATR model from two aspects. It is not enough to only compare the data level method, but the experiments also need to compare the algorithm-level method. Focal loss is an algorithm specially used to deal with imbalanced data. Therefore, it was also compared to the other results. Fig. 12 shows how the recognition performance of the various methods was affected when the proportion

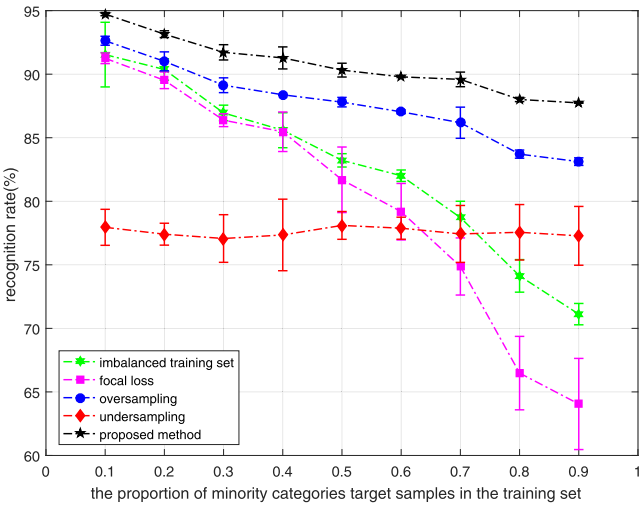


Fig. 12. Recognition performance of ATR model under the condition of different proportions of minority categories target samples.

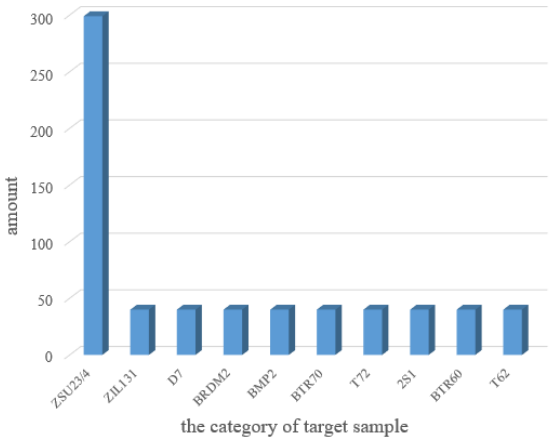


Fig. 13. Statistics of the training sample set when the proportion of the minority categories target samples reaches 90%.

of minority category target samples in the training set was gradually increased.

As shown in Fig. 12, the performance of the ATR model dropped more significantly as the proportion of the minority category target samples in the training set gradually increased. Among them, the recognition performance of the undersampling method was hardly affected, which is the reason why the degree of undersampling is closely related to the IR. Focal loss can only be effective when the proportion of minority samples does not exceed 50%, and its improvement effect is not as good as oversampling. At the same time, under different proportions of minority category target samples, CA-ATR demonstrated its advantages in dealing with imbalanced data. Moreover, the superiority of the proposed method became more apparent with the gradual increase in this proportion.

The statistics of the training sample set when this proportion of the MSTAR data set reached 90% are shown in Fig. 13. Such statistic features present a standard long-tailed distribution. At this proportion, the proposed method increased the baseline recognition rate of the original ATR

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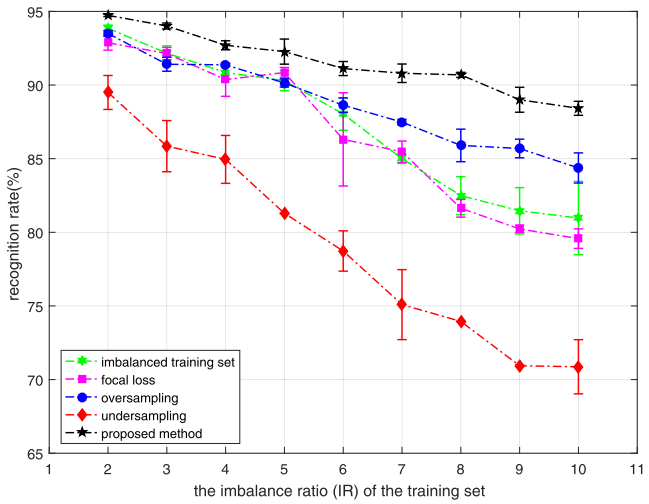


Fig. 14. Recognition performance of ATR model under the condition of different IR.

model by 15.74%. This verifies the ability of the proposed method to solve the long-tailed distribution data. Moreover, according to the variance of the recognition performance of various methods under different proportions, the recognition performance of the proposed method was also the most stable.

Additionally, the IR of minority category target samples was also a major factor affecting the recognition performance of ATR models. Therefore, the second experiment evaluated the impact of the IR on the above five situations. In the second experiment, the proportion of minority category target samples in the training set was fixed to 30% (there are three minority categories in the training set). The averages and variances of the experimental results through multiple experiments were recorded. Fig. 14 shows the recognition performance of the various methods when the IR of minority category target samples in the training set gradually increased.

The undersampling method is a way to solve the imbalanced features of minority category target samples. However, as shown in Fig. 14, this method gradually failed as the IR increased. When the IR did not exceed six, the recognition performance of focal loss was higher than that of oversampling. However, the recognition performance of focal loss was not stable. Moreover, as the IR continued to increase, focal loss gradually failed. On the contrary, the oversampling method remained effective as the IR gradually increased. When the IR of the training set exceeded sevens, the oversampling method effectively improved the overall recognition performance of the imbalanced training set. However, the superiority of the CA-ATR became more apparent with the gradual increase of the IR. When the IR of the training set reached ten, the proposed method increased the baseline recognition rate of the original ATR model by 5.65%. Similarly, the proposed method was able to remain stable under different IRs.

V. Conclusion

Imbalanced training data are often incompatible with data-driven ATR algorithms. This article proposed a CA-ATR model to deal with imbalanced data. This article gives a series of explanations on the derivation, architecture, and training details of the CA-ATR model. First, the detailed derivation of the loss function of cost-sensitive CNN was presented. Second, within the context of the two problems caused by imbalanced data sets, the intent of the corresponding architecture was explained in detail. Finally, the unique iterative optimization strategy for cost-sensitive awareness was also illustrated in this article. We also implement a series of experiments, which demonstrated the superiority of the proposed method. When faced with the two problems caused by imbalanced data, the two components of the proposed method, respectively, demonstrated their ability to address the corresponding problems. Moreover, the proposed method was able to solve different types of imbalanced data problems. When the proportion of the minority category target samples of the training data set reached 90%, the proposed method increased the baseline recognition rate of the original ATR model by 15.74%. At the same time, when the IR of the training set reached ten, the proposed method also increased the baseline recognition rate of the original ATR model by 5.65%. Overall, this article verified that the proposed CA-ATR model is an effective ATR model for imbalanced data, demonstrated its accuracy and application. The proposed method also provides a foundation for solving the problem of imbalanced data in ATR.

References

[1] Y. Sun, Z. Liu, S. Todorovic, and J. Li, "Adaptive boosting for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 1, pp. 112–125, Jan. 2007.

[2] Z. Jianxiong, S. Zhiguang, C. Xiao, and F. Qiang, "Automatic target recognition of SAR images based on global scattering center model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3713–3729, Oct. 2011.

[3] J. A. O'Sullivan, M. D. DeVore, V. Kedia, and M. I. Miller, "SAR ATR performance using a conditionally Gaussian model," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 1, pp. 91–108, Jan. 2001.

[4] U. Srinivas, V. Monga, and R. G. Raj, "SAR automatic target recognition using discriminative graphical models," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 1, pp. 591–606, Jan. 2014.

[5] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis with a new tensor nuclear norm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 925–938, Apr. 2020.

[6] X. Deng, X. Tian, S. Chen, and C. J. Harris, "Nonlinear process fault diagnosis based on serial principal component analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 560–572, Mar. 2018.

[7] J. Yang, D. Zhang, A. F. Frangi, and J.-Y. Yang, "Two-dimensional pca: A new approach to appearance-based face representation and recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 1, pp. 131–137, Jan. 2004.

[8] Z. Cui, Z. Cao, J. Yang, J. Feng, and H. Ren, "Target recognition in synthetic aperture radar images via non-negative matrix factorisation," *IET Radar, Sonar Navigat.*, vol. 9, no. 9, pp. 1376–1385, Dec. 2015.

[9] T. Cooke, "Two variations on Fisher's linear discriminant for pattern recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 2, pp. 268–273, 2002.

[10] S. Pang, S. Ozawa, and N. Kasabov, "Incremental linear discriminant analysis for classification of data streams," *IEEE Trans. Syst., Man Cybern., B, Cybern.*, vol. 35, no. 5, pp. 905–914, Oct. 2005.

[11] C. Lardeux *et al.*, "Support vector machine for multifrequency SAR polarimetric data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 12, pp. 4143–4152, Dec. 2009.

[12] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.

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[13] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.
[14] J. Pei, Y. Huang, W. Huo, Y. Zhang, J. Yang, and T.-S. Yeo, "SAR automatic target recognition based on multiview deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2196–2210, Apr. 2018.
[15] Y. Zhou, H. Wang, F. Xu, and Y.-Q. Jin, "Polarimetric SAR image classification using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1935–1939, Dec. 2016.
[16] X. Liu, Y. Huang, J. Pei, and J. Yang, "Sample discriminant analysis for SAR ATR," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 12, pp. 2120–2124, Dec. 2014.
[17] J. Choe, S. Park, K. Kim, J. H. Park, D. Kim, and H. Shim, "Face generation for low-shot learning using generative adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2017, pp. 1940–1948.
[18] J. Guo, B. Lei, C. Ding, and Y. Zhang, "Synthetic aperture radar image synthesis by using generative adversarial nets," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 7, pp. 1111–1115, Jul. 2017.
[19] C. Cao, Z. Cao, and Z. Cui, "LDGAN: A synthetic aperture radar image generation method for automatic target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3495–3508, May 2020.
[20] D. Malmgren-Hansen, A. Kusk, J. Dall, A. A. Nielsen, R. Engholm, and H. Skriver, "Improving SAR automatic target recognition models with transfer learning from simulated data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1484–1488, Sep. 2017.
[21] Z. Huang, Z. Pan, and B. Lei, "Transfer learning with deep convolutional neural network for SAR target classification with limited labeled data," *Remote Sens.*, vol. 9, no. 9, p. 907, Aug. 2017.
[22] S. H. Khan, M. Hayat, M. Bennamoun, R. Togneri, and F. A. Sohel, "A discriminative representation of convolutional features for indoor scene recognition," *IEEE Trans. Image Process.*, vol. 25, no. 7, pp. 3372–3383, Jul. 2016.
[23] S. H. Khan, M. Hayat, M. Bennamoun, F. A. Sohel, and R. Togneri, "Cost-sensitive learning of deep feature representations from imbalanced data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3573–3587, Aug. 2018.
[24] Z. Zheng, Y. Cai, and Y. Li, "Oversampling method for imbalanced classification," *Comput. Inform.*, vol. 34, no. 5, pp. 1017–1037, 2016.
[25] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.
[26] X.-Y. Jing *et al.*, "Multiset feature learning for highly imbalanced data classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 1, pp. 139–156, Jan. 2021.
[27] C. Huang, Y. Li, C. C. Loy, and X. Tang, "Learning deep representation for imbalanced classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5375–5384.
[28] K.-H. Huang and H.-T. Lin, "Cost-sensitive label embedding for multi-label classification," *Mach. Learn.*, vol. 106, nos. 9–10, pp. 1725–1746, Oct. 2017.
[29] C. L. Castro and A. P. Braga, "Novel cost-sensitive approach to improve the multilayer perceptron performance on imbalanced data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 24, no. 6, pp. 888–899, Jun. 2013.
[30] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *J. Big Data*, vol. 6, no. 1, p. 60, Dec. 2019.
[31] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.
[32] C. Bunkhumpornpat, K. Sinapiromsaran, and C. Lursinsap, "Safe-level-SMOTE: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem," in *Proc. Pacific–Asia Conf. Knowl. Discovery Data Mining*. Berlin, Germany: Springer, 2009, pp. 475–482.
[33] T. Maciejewski and J. Stefanowski, "Local neighbourhood extension of SMOTE for mining imbalanced data," in *Proc. IEEE Symp. Comput. Intell. Data Mining (CIDM)*, Apr. 2011, pp. 104–111.
[34] P. Jeatrakul, K. W. Wong, and C. C. Fung, "Classification of imbalanced data by combining the complementary neural network and SMOTE algorithm," in *Proc. Int. Conf. Neural Inf. Process.* Berlin, Germany: Springer, Nov. 2010, pp. 152–159.
[35] E. Ramentol, Y. Caballero, R. Bello, and F. Herrera, "SMOTE-RSB: A hybrid preprocessing approach based on oversampling and undersampling for high imbalanced data-sets using SMOTE and rough sets theory," *Knowl. Inf. Syst.*, vol. 33, no. 2, pp. 245–265, Nov. 2012.
[36] N. Merkle, S. Auer, R. Müller, and P. Reinartz, "Exploring the potential of conditional adversarial networks for optical and SAR image matching," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 1811–1820, Jun. 2018.
[37] F. Gao, Y. Yang, J. Wang, J. Sun, E. Yang, and H. Zhou, "A deep convolutional generative adversarial networks (DCGANs)-based semi-supervised method for object recognition in synthetic aperture radar (SAR) images," *Remote Sens.*, vol. 10, no. 6, p. 846, May 2018.
[38] C. P. Schwegmann, W. Kleynhans, B. P. Salmon, L. W. Mdakane, and R. G. V. Meyer, "Synthetic aperture radar ship discrimination, generation and latent variable extraction using information maximizing generative adversarial networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 2263–2266.
[39] S. Wang, W. Liu, J. Wu, L. Cao, Q. Meng, and P. J. Kennedy, "Training deep neural networks on imbalanced data sets," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2016, pp. 4368–4374.
[40] K. Nemoto, R. Hamaguchi, T. Imaizumi, and S. Hikosaka, "Classification of rare building change using CNN with multi-class focal loss," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 4663–4666.
[41] E. R. Keydel, S. W. Lee, and J. T. Moore, "MSTAR extended operating conditions: A tutorial," in *Algorithms for Synthetic Aperture Radar Imagery III*, vol. 2757. International Society for Optics and Photonics, Jun. 1996, pp. 228–242.



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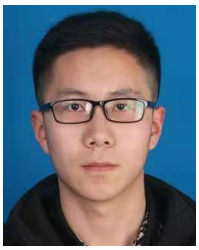
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